Chapter 2: Complex Analysis

Special Functions of Mathematical Physics: A Tourist's Guidebook

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Outline

Where we are going

To analyze the special functions of physics we first build the tools of *complex analysis*.

- **Taylor and Laurent series** represent a function near a point, even near a singularity.
- **Analytic functions** (those with a derivative in the complex sense) obey the *Cauchy–Riemann conditions* and are infinitely differentiable.
- **Cauchy's integral theorem** and the **residue theorem** turn contour integrals into a count of poles, giving a powerful machine for evaluating real integrals and infinite sums.
- The **method of steepest descent** extracts the large-parameter behavior of integrals—the asymptotics we will need for the named functions.

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Matching a polynomial to a function

We want the best local polynomial approximation

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_n x^n$$

to a smooth function $f(x)$ near $x = 0$. Demand that p and its derivatives agree with those of f at the origin: $f(0) = p(0)$, $f'(0) = p'(0)$, $f''(0) = p''(0)$, and so on. Differentiating p and evaluating at $x = 0$ strips off one coefficient at a time, because every term with a positive power of x vanishes there:

$$p(0) = a_0 \;\Rightarrow\; a_0 = f(0), \qquad p'(0) = a_1 \;\Rightarrow\; a_1 = f'(0),$$
$$p''(0) = 2a_2 \;\Rightarrow\; a_2 = \tfrac{1}{2} f''(0), \qquad p^{(3)}(0) = 6a_3 \;\Rightarrow\; a_3 = \tfrac{1}{6} f^{(3)}(0),$$

and in general $p^{(n)}(0) = n!\, a_n$ gives

$$a_n = \frac{1}{n!} f^{(n)}(0). \tag{2.1.1}$$

Here $n! = n \cdot (n-1) \cdots 2 \cdot 1$ is the factorial.

Matching a polynomial to a function (cont.)

Collecting the coefficients gives the *Taylor polynomial*. Letting the number of terms $n \to \infty$ gives the *formal Taylor series*; when f is analytic, this series converges back to f in its interval (or disk) of convergence:

$$f(x) = f(0) + xf'(0) + \tfrac{1}{2}x^2 f''(0) + \tfrac{1}{3!}x^3 f'''(0) + \cdots\,, \tag{2.1.2}$$

for x in that convergence region. About an arbitrary point x_0,

$$f(x) = f(x_0) + (x - x_0)f'(x_0) + \tfrac{1}{2}(x - x_0)^2 f''(x_0) + \tfrac{1}{3!}(x - x_0)^3 f'''(x_0) + \cdots\,. \tag{2.1.3}$$

A function that has a Taylor series representation at x_0 is said to be *analytic* at x_0.

Examples: the elementary power series

Exponential. Define e^x by $\frac{d}{dx}e^x = e^x$ with $e^0 = 1$. Then $f^{(n)}(0) = 1$ for all n, so (2.1.2) gives

$$e^x = 1 + x + \tfrac{1}{2}x^2 + \tfrac{1}{3!}x^3 + \tfrac{1}{4!}x^4 + \cdots . \tag{2.1.4}$$

Since $n!$ grows faster than x^n, the series converges for all x.

Cosine and sine. Using $\frac{d}{dx}\cos x = -\sin x$ and $\frac{d}{dx}\sin x = \cos x$, together with $\sin 0 = 0$, $\cos 0 = 1$,

$$\cos x = 1 - \tfrac{1}{2}x^2 + \tfrac{1}{4!}x^4 - \cdots , \tag{2.1.5}$$

$$\sin x = x - \tfrac{1}{3!}x^3 + \tfrac{1}{5!}x^5 - \cdots . \tag{2.1.6}$$

Both converge for all x.

Detour: arithmetic as repeated operations

The book's arithmetic detour puts familiar operations into one ladder:

$$\text{succession} \longrightarrow \text{addition} \longrightarrow \text{multiplication} \longrightarrow \text{exponentiation} \longrightarrow \text{tetration.}$$

Each operation repeats the previous one:

$$x + y = \underbrace{S(S(\cdots S(x)))}_{y \ \text{times}}, \qquad xy = \underbrace{x + x + \cdots + x}_{y \ \text{times}}, \qquad x^y = \underbrace{x \cdot x \cdots x}_{y \ \text{times}}.$$

The next operation, *tetration*, repeats exponentiation. Let $T_1(x) = x$ and $T_{y+1}(x) = x^{T_y(x)}$. Then

$$T_1(3) = 3, \qquad T_2(3) = 3^3 = 27, \qquad T_3(3) = 3^{27} = 7{,}625{,}597{,}484{,}987.$$

The book also writes tetration as $x \uparrow\uparrow y$, or as a left superscript in some older notation. The next value

$$T_4(3) = 3^{3^{27}}$$

has

$$\left\lfloor 3^{27} \log_{10} 3 \right\rfloor + 1 \approx 3.64 \times 10^{12}$$

decimal digits. This is the point of the detour: iterated operations quickly outgrow ordinary notation and force us to invent better symbolic machinery.

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Imaginary unit, Cartesian and polar form

Define the imaginary unit i by $i^2 = -1$, i.e. $i = \sqrt{-1}$. A *complex number* is a linear combination of real and imaginary parts. Powers of i cycle:

$$\{i^0, i^1, i^2, i^3, i^4, \dots\} = \{1, i, -1, -i, 1, \dots\}.$$

In *Cartesian* form $z = x + iy$; we call $x = \operatorname{Re} z$ the real part and $y = \operatorname{Im} z$ the imaginary part.
Put $x = i\theta$ in the exponential series (2.1.4). Each power of i follows the cycle $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, so

$$e^{i\theta} = 1 + i\theta + \tfrac{1}{2}(i\theta)^2 + \tfrac{1}{3!}(i\theta)^3 + \tfrac{1}{4!}(i\theta)^4 + \cdots = 1 + i\theta - \tfrac{1}{2}\theta^2 - i\tfrac{1}{3!}\theta^3 + \tfrac{1}{4!}\theta^4 + \cdots .$$

The even powers are real and the odd powers carry a factor i; grouping them and comparing with (2.1.5)–(2.1.6) gives *Euler's formula*:

$$e^{i\theta} = \left(1 - \tfrac{1}{2}\theta^2 + \tfrac{1}{4!}\theta^4 - \cdots\right) + i\left(\theta - \tfrac{1}{3!}\theta^3 + \tfrac{1}{5!}\theta^5 - \cdots\right) = \cos\theta + i\sin\theta. \tag{2.2.1}$$

Imaginary unit, Cartesian and polar form (cont.)

Euler's formula gives the *polar* form of a complex number,

$$z = r\,e^{i\theta}, \tag{2.2.2}$$

where $r = |z|$ is the *modulus* (magnitude) and $\theta = \arg z$ is the *argument* (phase). From $z = x + iy$,

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \arg z,$$

where $\arg z$ means the quadrant-aware angle (often computed as $\operatorname{atan2}(y, x)$), not the single-valued real arctangent alone. Multiplication scales by the moduli and adds the arguments: if $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ then $z_1 z_2 = r_1 r_2\, e^{i(\theta_1+\theta_2)}$.
The *complex conjugate* of $z = re^{i\theta}$ is $z^* = re^{-i\theta}$ (in Cartesian form $(x + iy)^* = x - iy$), so that

$$zz^* = \left(re^{i\theta}\right)\left(re^{-i\theta}\right) = r^2 = |z|^2.$$

The conjugate also recovers the Cartesian components:

$$\operatorname{Re} z = \frac{z + z^*}{2}, \qquad \operatorname{Im} z = \frac{z - z^*}{2i}. \tag{2.2.3}$$

Imaginary unit, Cartesian and polar form (cont.)

From Euler's formula, $e^{i\theta} = \cos\theta + i\sin\theta$ and (replacing θ by $-\theta$, since cosine is even and sine is odd) $e^{-i\theta} = \cos\theta - i\sin\theta$. Adding the two cancels the sines, while subtracting cancels the cosines:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \tag{2.2.4}$$

We will use these identities throughout.

Numerical derivatives and the complex step

A truncated Taylor series with a small step h estimates a derivative. From

$$f(x+h) = f(x) + hf'(x) + \tfrac{1}{2}h^2 f''(x) + \tfrac{1}{6}h^3 f'''(x) + \cdots,$$

solving for $f'(x)$ gives us

$$f'(x) = \frac{f(x+h) - f(x)}{h} - \tfrac{1}{2}hf''(x) - \tfrac{1}{6}h^2 f'''(x) + \cdots, \tag{2.2.5}$$

so the forward difference is first order: $f'(x) = \dfrac{f(x+h) - f(x)}{h} + O(h)$.
For the centered difference, write both Taylor expansions:

$$f(x+h) = f(x) + hf'(x) + \tfrac{1}{2}h^2 f''(x) + \tfrac{1}{6}h^3 f'''(x) + \cdots,$$
$$f(x-h) = f(x) - hf'(x) + \tfrac{1}{2}h^2 f''(x) - \tfrac{1}{6}h^3 f'''(x) + \cdots.$$

Subtracting cancels the even terms:

$$f(x+h) - f(x-h) = 2hf'(x) + \tfrac{1}{3}h^3 f'''(x) + \cdots.$$

Solving for $f'(x)$ gives the *centered difference*, which is second order:

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} - \tfrac{1}{6}h^2 f'''(x) + \cdots = \frac{f(x+h) - f(x-h)}{2h} + O(h^2). \tag{2.2.6}$$

Numerical derivatives and the complex step (cont.)

In floating point, $f(x+h) - f(x)$ is corrupted by round-off of size $|f(x)|\varepsilon$ (machine epsilon $\varepsilon \approx 10^{-16}$), so the forward difference carries two errors:

$$f'(x) = \frac{f(x+h) - f(x)}{h} \underbrace{+\, O(h)}_{\text{truncation}} \underbrace{+\, O\left(\frac{|f(x)|\varepsilon}{h}\right)}_{\text{round-off}}.$$

For a real, analytic f we can sidestep the round-off by stepping in the imaginary direction. From

$$f(x+ih) = f(x) + ihf'(x) - \tfrac{1}{2}h^2 f''(x) - i\tfrac{1}{6}h^3 f'''(x) + \cdots,$$

taking the imaginary part gives $\mathrm{Im}[f(x+ih)] = hf'(x) - \frac{1}{6}h^3 f'''(x) + \cdots$. Solving for $f'(x)$ gives us

$$f'(x) = \frac{\mathrm{Im}[f(x+ih)]}{h} + \tfrac{1}{6}h^2 f'''(x) + \cdots = \frac{\mathrm{Im}[f(x+ih)]}{h} + O(h^2). \tag{2.2.7}$$

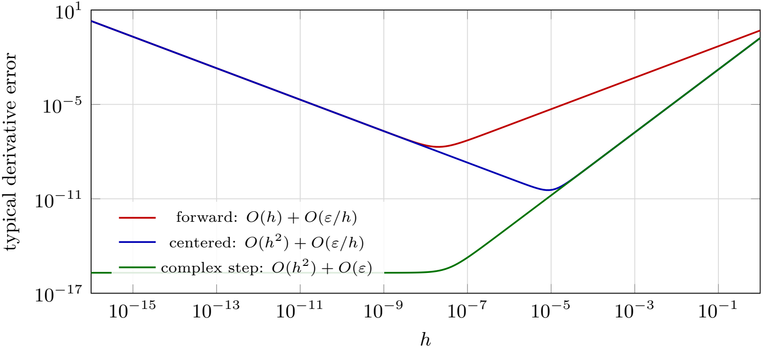
No subtraction of nearly equal numbers occurs, so this *complex-step* derivative is not limited by round-off error.

Derivative error as the step size changes

The Taylor expansion predicts the slopes of the error curves. For a typical smooth function in double precision,

$$E_{\text{forward}}(h) \sim Ch + \frac{C'\varepsilon}{h}, \qquad E_{\text{centered}}(h) \sim Ch^2 + \frac{C'\varepsilon}{h}, \qquad E_{\text{complex}}(h) \sim Ch^2 + C'\varepsilon.$$

The complex-step curve avoids the $1/h$ round-off wall because no nearly equal real numbers are subtracted.



Multivalued functions and branch cuts

A function of a real variable is easy to graph; its inverse may be *multivalued*. For $f(x) = x^2$ the inverse has two branches, $f^{-1}(y) = \pm\sqrt{y}$ for $y \geq 0$. We pick a *principal branch*, usually $f^{-1}(y) = +\sqrt{y}$, to make it single-valued.
The arcsine is also multivalued: every $x = n\pi$ maps to $y = 0$ (more generally x, $x + 2\pi n$, and $\pi - x$ share the same sine). Since $\sin n\pi = 0$ for all integers n, $\sin^{-1} 0 = n\pi$ for all integer n. We restrict to the principal branch $x \in [-\pi/2, \pi/2]$.

Multivalued functions and branch cuts (cont.)

The logarithm. As the inverse of the exponential, $\log z$ is multivalued. Writing $z = re^{i\theta}$ and inserting the ambiguity $e^{i2\pi n} = 1$,

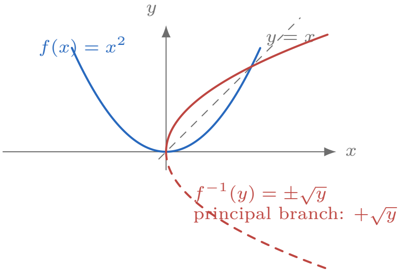
$$\begin{aligned}\log z &= \log r + \log e^{i\theta} = \log r + \log e^{i(\theta+2\pi n)} \\ &= \log r + i(\theta + 2\pi n) = \log|z| + i \arg z,\end{aligned}$$

where $\arg z = \theta + 2\pi n$. The principal branch takes $-\pi < \operatorname{Arg} z < \pi$:

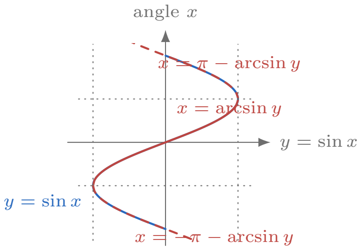
$\operatorname{Log} z = \log|z| + i \operatorname{Arg} z$ with $\operatorname{Arg} z$ chosen in that interval. To keep a multivalued function single-valued we make a *branch cut* in the plane. A natural cut for $\sqrt{z}$ or $\log z$ runs along the negative real axis, but any simple path from the branch point to infinity can be used if it keeps the important region continuous. Crossing the cut, $\log z$ jumps by $2\pi i$.

Real graphs and inverse branches

real function and inverse branches



$\sin x$, arcsin, and branch choices



principal branch uses $-\pi/2 \leq x \leq \pi/2$

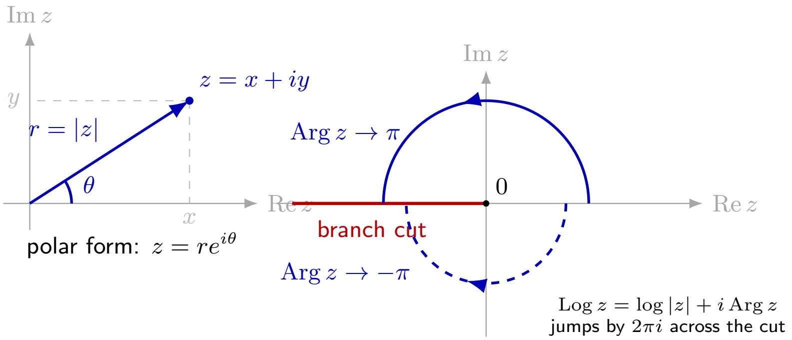
The book first motivates branch choices using real inverse functions: x^2 has two inverse branches, and $\sin x$ repeats values infinitely often. Complex roots and logarithms behave the same way, except the branch choice is made in the plane.

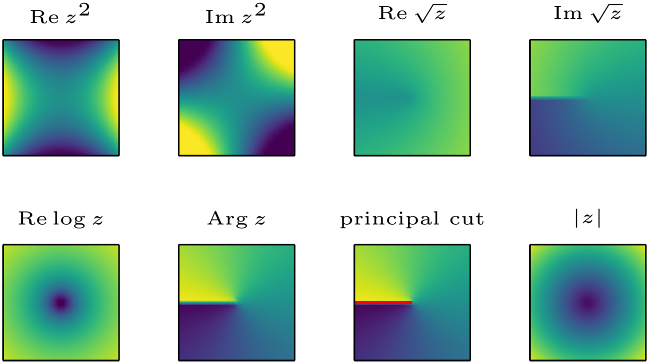
Complex-variable pictures and branch cuts

A complex-valued function has two real inputs and two real outputs:

$$z = x + iy, \qquad f(z) = u(x, y) + iv(x, y).$$

Instead of one graph, we often inspect level curves, real and imaginary parts, or how a chosen branch changes around a singular point.





For $f(z) = u(x, y) + iv(x, y)$, a single real graph is replaced by component plots. The gallery shows the real and imaginary parts of z^2, the principal square root, and the principal logarithm; the jump in $\operatorname{Arg} z$ is exactly the branch cut.

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The complex derivative and Cauchy–Riemann

The derivative of $f(z)$ is

$$f'(z) = \frac{df}{dz} = \lim_{\delta z \to 0} \frac{f(z+\delta z) - f(z)}{\delta z}, \tag{2.3.1}$$

provided the limit is the same for *all* directions δz. If the derivative exists only at z_0, f is complex differentiable there. If it exists in a neighborhood of z_0, f is *analytic* at z_0; if it exists everywhere in $\mathbb{C}$, f is *entire*; where it fails, z_0 is a *singular point*.

Write $z = x + iy$ and $f = u(x,y) + iv(x,y)$. Approaching along the real axis ($\delta z = \delta x$, $\delta y = 0$):

$$\frac{df}{dz} = \frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}.$$

Approaching along the imaginary axis ($\delta z = i\,\delta y$, $\delta x = 0$) divides by the extra factor i; using $1/i = -i$,

$$\frac{df}{dz} = \frac{1}{i}\frac{\partial f}{\partial y} = -i\left(\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right) = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}.$$

The complex derivative and Cauchy–Riemann (cont.)

For the derivative to exist these must agree. Matching real and imaginary parts gives the *Cauchy–Riemann conditions*:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}. \tag{2.3.2}$$

Equivalently, $i\,\partial f/\partial x = \partial f/\partial y$. These conditions are necessary for analyticity. If the first partial derivatives are continuous in a neighborhood, they are also sufficient, so in that common smooth case $f = u + iv$ is analytic if and only if it satisfies the Cauchy–Riemann conditions.

Examples.

- $z^2 = (x^2 - y^2) + i\,2xy$ is entire: $u_x = 2x = v_y$ and $u_y = -2y = -v_x$.
- On a chosen branch, $\log z$ is analytic away from 0 and the branch cut. Locally

  $$f(x, y) = \log\sqrt{x^2 + y^2} + i\ \mathrm{Arg}(x + iy),$$

  and

  $$\frac{\partial f}{\partial x} = \frac{x}{x^2 + y^2} - i\frac{y}{x^2 + y^2}, \qquad \frac{\partial f}{\partial y} = \frac{y}{x^2 + y^2} + i\frac{x}{x^2 + y^2},$$

  so $i\partial_x f = \partial_y f$ on that branch.

The complex derivative and Cauchy–Riemann (cont.)

- $1/z$ is analytic except at $z = 0$. With $f = 1/(x + iy)$, $\partial_x f = -1/(x + iy)^2$ and $\partial_y f = -i/(x + iy)^2$, so $i\partial_x f = \partial_y f$ except at the origin, where f blows up.
- $z^* = x - iy$ has $u = x$, $v = -y$, so $u_x = 1$ but $v_y = -1$: the Cauchy–Riemann conditions never hold, z^* is nowhere analytic.

A non-constant polynomial has a genuine *pole at infinity*; other entire functions need not (e.g. e^z has an essential singularity at ∞, and constants are bounded).

Harmonic functions, potentials, streamlines

Laplace's equation $\nabla^2 u = 0$ governs steady-state temperature, electrostatic potential, and elastic membranes. Its solutions are *harmonic functions*. If $f = u + iv$ is analytic, both u and v are harmonic. From the Cauchy–Riemann conditions,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial y\, \partial x}, \qquad \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial x\, \partial y},$$

and adding cancels the mixed partials:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \qquad \text{similarly} \;\; \nabla^2 v = 0. \tag{2.3.3}$$

A nonconstant harmonic function has no strict local maximum or minimum in the interior of its domain. Away from flat critical points, this is why its level curves look saddle-like rather than peak-like or bowl-like.

Harmonic functions, potentials, streamlines (cont.)

In fluid mechanics and electrostatics we call u the *complex potential* and v the *stream function*. Curves $u = \text{const}$ are *equipotentials*, curves $v = \text{const}$ are *streamlines*. Their gradients are orthogonal, since by Cauchy–Riemann

$$\nabla u \cdot \nabla v = \frac{\partial u}{\partial x}\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\frac{\partial v}{\partial y} = 0.$$

Example (z^2). With $f = z^2 = (x^2 - y^2) + i\,2xy$, the potential is $u = x^2 - y^2$ and the stream function is $v = 2xy$. Thus

$$u = c_1 \quad \Longleftrightarrow \quad x^2 - y^2 = c_1, \qquad v = c_2 \quad \Longleftrightarrow \quad 2xy = c_2.$$

These are two orthogonal families of hyperbolas. The orthogonality check is direct: $\nabla u = (2x, -2y)$, $\nabla v = (2y, 2x)$, so $\nabla u \cdot \nabla v = 4xy - 4xy = 0$.

Example ($1/z$). With $f = 1/z$ and $z = x + iy$,

$$f = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2}.$$

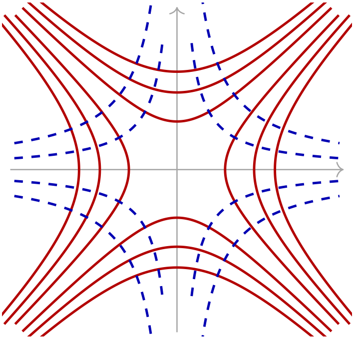
Harmonic functions, potentials, streamlines (cont.)

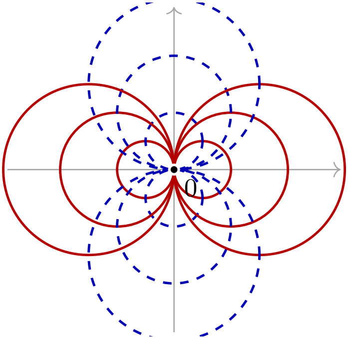
In polar coordinates ($x = r\cos\theta$, $y = r\sin\theta$),

$$u(r,\theta) + iv(r,\theta) = r^{-1}\cos\theta - i\,r^{-1}\sin\theta,$$

so $u = r^{-1}\cos\theta$ is the complex potential and $v = -r^{-1}\sin\theta$ is the stream function. The equation $r^{-1}\cos\theta = c$ rearranges to $r = (1/c)\cos\theta$; after renaming $1/c$ as c_1, the equipotentials are $r = c_1\cos\theta$. Similarly the streamlines are $r = c_2\sin\theta$.

Harmonic functions, potentials, streamlines (cont.)





— $u =$ constant - - - $v =$ constant

Harmonic functions, potentials, streamlines (cont.)

The red curves are $u =$ constant and the dashed blue curves are $v =$ constant. The Cauchy–Riemann equations force the two families to meet orthogonally wherever $f'(z) \neq 0$.

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Contour integrals and a key circle integral

Along the real axis, the usual Riemann integral is

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{j=0}^{n-1} f(x_j)(x_{j+1} - x_j).$$

For a path γ from $z = c$ to $z = d$, the same Riemann-sum idea defines the *path* (contour, line) integral:

$$\int_\gamma f(z)\,dz = \lim_{n\to\infty} \sum_{i=0}^{n-1} f(z_i)\,(z_{i+1} - z_i), \tag{2.4.1}$$

with $z_0 = c$, $z_n = d$. If $c = d$ the path is closed, written $\oint_\gamma$; by convention closed paths are traversed counterclockwise.

Powers of z on a circle. On $|z| = R$ write $z = Re^{i\theta}$, $dz = iRe^{i\theta}\,d\theta$. Then

$$\oint_{|z|=R} z^n\,dz = iR^{n+1} \int_0^{2\pi} e^{i(n+1)\theta}\,d\theta = \begin{cases} 2\pi i, & n = -1, \\ 0, & n \neq -1. \end{cases}$$

Contour integrals and a key circle integral (cont.)

(For $n = -1$ the integrand is $i\,d\theta$; for $n \neq -1$ the exponential integrates to a periodic function over a full period.) In general,

$$\oint_{|z-z_0|=R} (z - z_0)^n \, dz = 2\pi i \, \delta_{n,-1}. \tag{2.4.2}$$

Contour integrals and a key circle integral (cont.)

Antiderivatives. Parameterize γ from $z = a$ to $z = b$ as $z = \gamma(t)$, $t \in [0, 1]$, with $\gamma(0) = a$ and $\gamma(1) = b$. Let $F' = f$. By the fundamental theorem of calculus,

$$\int_\gamma f(z)\,dz = \int_0^1 F'(\gamma(t))\gamma'(t)\,dt = F(\gamma(1)) - F(\gamma(0)) = F(b) - F(a).$$

If γ is closed ($a = b$) and F is single-valued, then $\oint_\gamma f\,dz = 0$.
Reworking the circle example over any simple closed γ around the origin:

- $n \neq -1$: $F(z) = z^{n+1}/(n+1)$ is single-valued, so $\oint_\gamma z^n\,dz = 0$.
- $n = -1$: $F(z) = \log z$ has a branch cut. Put the cut on the negative real axis and shorten γ by $\pm\varepsilon$ on each side. On the cut-open curve, use $\log z = \log|z| + i \arg z$: the radius returns to its starting value while the angle runs from $\theta = -\pi + \varepsilon$ to $\theta = \pi - \varepsilon$, so

  $$\int_\gamma \frac{1}{z}\,dz = \Big[\log r + i\theta\Big]_{\theta=-\pi+\varepsilon}^{\theta=\pi-\varepsilon} = i(2\pi - 2\varepsilon) = 2\pi i - 2\varepsilon i.$$

  As $\varepsilon \to 0$, $\oint_\gamma \frac{1}{z}\,dz = 2\pi i$.

Cauchy's integral theorem and formula

Theorem (Cauchy's Integral Theorem)

If $f(z)$ is analytic in a simply connected region and γ is a closed path in that region, then $\oint_\gamma f(z)\,dz = 0$.

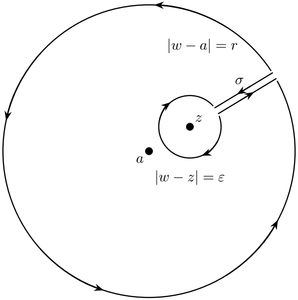
Proof. The following calculation proves the smooth case; the full Cauchy–Goursat theorem removes this extra smoothness assumption. Assume first that $\gamma = \partial D$ is positively oriented, piecewise smooth, and simple. With $f = u + iv$ and $dz = dx + i\,dy$, Green's theorem and Cauchy–Riemann give

$$\begin{aligned}\oint_\gamma f\,dz &= \oint_\gamma (u\,dx - v\,dy) + i\oint_\gamma (v\,dx + u\,dy)\\ &= \iint_D \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) dx\,dy + i\iint_D \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dx\,dy = 0,\end{aligned}$$

since each integrand vanishes by (2.3.2). □

Cauchy's integral theorem and formula (cont.)

Cauchy's integral formula. Let f be analytic and γ encircle z_0. Deform γ to a small circle $z = z_0 + \varepsilon e^{i\theta}$; the two sides of the connecting slit are traversed in opposite directions, so they cancel:



Cauchy's integral theorem and formula (cont.)

On the small circle, $z - z_0 = \varepsilon e^{i\theta}$ and $dz = i\varepsilon e^{i\theta}\, d\theta$, so the factor $\varepsilon e^{i\theta}$ cancels between numerator and denominator:

$$\oint_\gamma \frac{f(z)}{z - z_0}\, dz = \int_0^{2\pi} \frac{f(z_0 + \varepsilon e^{i\theta})}{\varepsilon e^{i\theta}}\, i\varepsilon e^{i\theta}\, d\theta = i \int_0^{2\pi} f(z_0 + \varepsilon e^{i\theta})\, d\theta.$$

As $\varepsilon \to 0$ continuity gives $f(z_0 + \varepsilon e^{i\theta}) \to f(z_0)$, a constant that comes out of the integral, and $\int_0^{2\pi} d\theta = 2\pi$:

$$\oint_\gamma \frac{f(z)}{z - z_0}\, dz = i f(z_0) \int_0^{2\pi} d\theta = 2\pi i\, f(z_0). \tag{2.4.3}$$

Differentiating Cauchy's formula with respect to z_0 (equivalently, matching the Taylor coefficient of f at z_0) gives the formula for higher poles and the *derivative formula*. The first formula is for $n = 1, 2, \ldots$; the second is for $n = 0, 1, 2, \ldots$:

$$\oint_\gamma \frac{f(z)}{(z - z_0)^n}\, dz = \frac{2\pi i}{(n-1)!}\, f^{(n-1)}(z_0), \qquad f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}}\, dz. \tag{2.4.4}$$

Cauchy's integral theorem and formula (cont.)

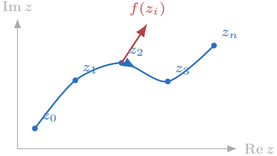
Orientation check. Let

$$I = \oint_\gamma \frac{z^2}{z-2}\,dz.$$

If γ winds counterclockwise once around $z = 2$, Cauchy's formula gives $I = 2\pi i\, f(2)$ with $f(z) = z^2$, hence $I = 2\pi i \cdot 4 = 8\pi i$. If the same curve is traversed clockwise, the orientation reverses the sign: $I = -8\pi i$. If γ does not enclose $z = 2$, the integrand is analytic inside γ, so Cauchy's theorem gives $I = 0$.

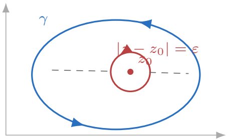
Geometry behind contour arguments

path integral as a Riemann sum



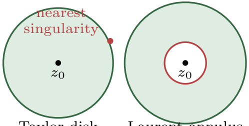
mesh the path, multiply by
the complex increments Δz_i

Cauchy formula: shrink to z_0



connecting slits cancel;
only the small circle remains

Taylor disk vs Laurent annulus



Taylor disk
$|z - z_0| < R$

Laurent annulus
$r < |z - z_0| < R$

The same three pictures recur throughout the chapter: Riemann sums along a path, Cauchy's deformation to a small circle around z_0, and the disk/annulus distinction between Taylor and Laurent convergence.

Worked example: partial fractions and Cauchy

Evaluate $\displaystyle\oint_{|z|=2} \frac{e^z}{z^2(z-1)}\,dz.$

Partial fractions: $\displaystyle\frac{1}{z^2(z-1)} = \frac{az+b}{z^2} + \frac{c}{z-1}$. Clearing denominators, $(az+b)(z-1)+cz^2 = 1$, so $az^2+bz-az-b+cz^2 = 1$. Matching coefficients,

$$\left.\begin{aligned} a+c&=0\\ b-a&=0\\ -b&=1 \end{aligned}\right\} \Rightarrow\ c=1,\quad a=-1,\quad b=-1.$$

Hence the integrand splits as

$$\frac{e^z}{z^2(z-1)} = \frac{e^z}{z-1} - \frac{(z+1)e^z}{z^2}.$$

Worked example: partial fractions and Cauchy (cont.)

The first piece: by Cauchy's integral formula (2.4.3) with $z_0 = 1$ inside $|z| = 2$,

$$\oint_{|z|=2} \frac{e^z}{z-1}\, dz = 2\pi i\, e^1.$$

The second piece: set $f(z) = (z+1)e^z$, a double pole at $z = 0$. Then

$$f'(z) = e^z + (z+1)e^z = (z+2)e^z, \qquad f'(0) = 2,$$

so by the derivative formula (2.4.4) (with $n = 2$), $\oint_{|z|=2} \frac{(z+1)e^z}{z^2}\, dz = 2\pi i\, f'(0) = 2\pi i(2)$. Therefore

$$\oint_{|z|=2} \frac{e^z}{z^2(z-1)}\, dz = 2\pi i(e-2).$$

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Taylor series from contour integration

Theorem (Taylor Series)

A function $f(z)$ analytic in a neighborhood of z_0 has $f(z) = \displaystyle\sum_{n=0}^{\infty} \frac{1}{n!}(z-z_0)^n f^{(n)}(z_0)$.

Proof. Start from Cauchy's formula and expand the kernel as a geometric series. With $|z - z_0| < |\xi - z_0|$,

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \oint \frac{f(\xi)}{\xi - z}\, d\xi = \frac{1}{2\pi i} \oint \frac{f(\xi)}{\xi - z_0} \frac{1}{1 - \frac{z-z_0}{\xi - z_0}}\, d\xi \\ &= \frac{1}{2\pi i} \oint \frac{f(\xi)}{\xi - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0}\right)^n d\xi = \frac{1}{2\pi i} \sum_{n=0}^{\infty} (z - z_0)^n \oint \frac{f(\xi)}{(\xi - z_0)^{n+1}}\, d\xi \\ &= \sum_{n=0}^{\infty} \frac{1}{n!}(z - z_0)^n f^{(n)}(z_0), \end{aligned}$$

using the derivative formula (2.4.4) in the last step. □

A function is analytic at z_0 if and only if it has a Taylor series there.

Taylor series from contour integration (cont.)

The geometric-series step uses $1 + r + r^2 + \cdots = \dfrac{1}{1-r}$ for $|r| < 1$, which follows from

$$(1-r)(1 + r + \cdots + r^n) = 1 - r^{n+1} \xrightarrow{n\to\infty} 1.$$

Binomial expansion. For $f(z) = (1+z)^m$ with real m, the nth derivative at $z = 0$ is

$$\frac{d^n}{dz^n}(1+z)^m\bigg|_{z=0} = m(m-1)\cdots(m-n+1),$$

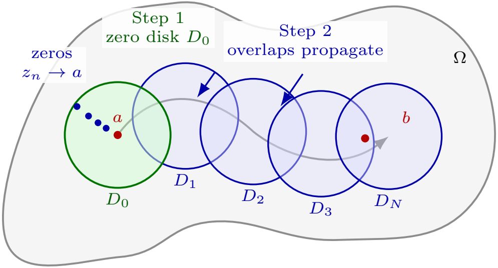
so

$$(1+z)^m = 1 + mz + \frac{m(m-1)}{1\cdot 2}z^2 + \cdots = \sum_{n=0}^{\infty}\binom{m}{n}z^n. \tag{2.5.1}$$

If m is a non-negative integer the series terminates. Otherwise it is infinite, and since $(1+z)^{1/2} = e^{\frac{1}{2}\log(1+z)}$ has a branch point at $z = -1$, the expansion about 0 converges only for $|z| < 1$.

Analytic continuation

A general smooth function on $[0,1]$ has no unique extension to $\mathbb{R}$, but a polynomial of degree n is fixed by $n+1$ sampled points. A Taylor series likewise has a *unique* extension wherever continuation is possible: in $\mathbb{C}$, an analytic function has a unique *analytic continuation* along overlapping domains until a singularity or branch obstruction is met.



Analytic continuation (cont.)

Example. The real graphs of $(1+x^2)^{-1}$ and e^{-x^2} look alike, but their Taylor series differ: e^{-x^2} converges everywhere, while the Taylor series of $(1+x^2)^{-1}$ about 0 has radius 1 and fails at $|x|=1$. In the plane,

$$\frac{1}{1+z^2} = \frac{1}{(i+z)(-i+z)}$$

has poles at $z=\pm i$, so the Taylor series about 0 converges only for $|z|<1$. The function e^{-z^2} is entire, so its series is valid everywhere.

Analytic continuation (cont.)

Theorem (Identity / splicing)

If f_1 and f_2 are analytic on a connected domain $U \cup V$, and if $f_1 = f_2$ on an overlap $U \cap V$ containing an accumulation point, then $f_1 = f_2$ on all of $U \cup V$.

Proof. Let $g = f_1 - f_2$. Then g is analytic on $U \cup V$ and $g = 0$ on $U \cap V$. Its Taylor series is identically zero there, and by uniqueness of analytic continuation $g \equiv 0$ on $U \cup V$. Hence $f_1 = f_2$. $\square$

Laurent series

Near a singularity a *Laurent series* beats a Taylor series. It splits f into analytic and singular parts:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n = \underbrace{\sum_{n=0}^{\infty} a_n (z - z_0)^n}_{\text{analytic}} + \underbrace{\sum_{n=-\infty}^{-1} a_n (z - z_0)^n}_{\text{singular}}, \qquad (2.5.2)$$

valid in an *annulus* about z_0 even when z_0 is singular. Multiplying by $(z - z_0)^{-k-1}$ and integrating term by term, only the $n = k$ term survives by (2.4.2), giving the coefficients

$$a_n = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}} \, dz. \qquad (2.5.3)$$

Laurent series (cont.)

Example (e^{-1/z^2}). On the real axis every derivative at 0 is zero, so the Taylor series is 0 everywhere—wrong off the origin, because $z = 0$ is an essential singularity. From the entire series for e^z,

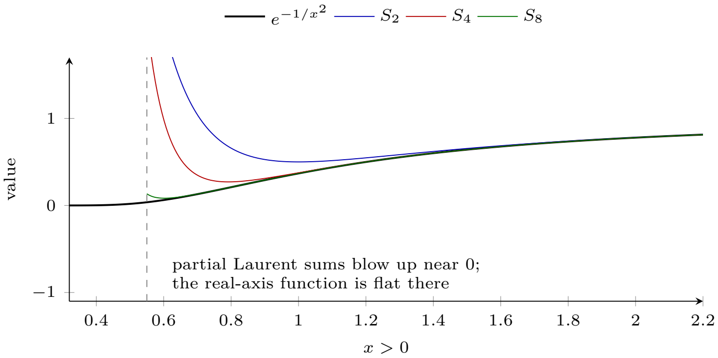
$$e^{-z^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} z^{2n} \;\Rightarrow\; e^{-1/z^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} z^{-2n},$$

a Laurent series valid for all $z \neq 0$.
The real-axis flatness follows from the standard limit

$$\lim_{x \to 0} x^{-m} e^{-1/x^2} = 0 \qquad (m = 0, 1, 2, \ldots).$$

Thus $f(0) = 0$, $f'(0) = \lim_{x\to 0} e^{-1/x^2}/x = 0$, and the same argument applies after each differentiation: every derivative is a polynomial in $1/x$ times e^{-1/x^2}, so every derivative at 0 vanishes.

Laurent series (cont.)



Laurent series (cont.)

Example (sinc). Since $\sin z = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1}$,

$$\operatorname{sinc} z = \frac{\sin z}{z} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n}.$$

The singularity at $z = 0$ is removable: defining $\operatorname{sinc} 0 = 1$ makes it analytic.

Laurent series and the Bernoulli numbers

Example ($1/(e^z - 1)$). This has a simple pole at 0. Work with $g(z) = \dfrac{e^z - 1}{z} = \sum_{j=0}^{\infty} \dfrac{1}{(j+1)!} z^j$ and invert it. The *Cauchy product* of two power series is

$$\Big(\sum_i a_i z^i\Big)\Big(\sum_j b_j z^j\Big) = \sum_k c_k z^k, \qquad c_k = \sum_{l=0}^{k} a_l b_{k-l},$$

so with $b_j = 1/(j+1)!$ and $c_0 = 1$, $c_k = 0$ ($k \geq 1$), the reciprocal coefficients satisfy $a_n = b_0^{-1}\big(c_n - \sum_{l=0}^{n-1} a_l b_{n-l}\big)$. The first few are

$$a_0 = 1, \quad a_1 = -\tfrac{1}{2}, \quad a_2 = \tfrac{1}{12}, \quad a_3 = 0,$$

$$a_4 = -\Big(\tfrac{1}{120}\cdot 1 + \tfrac{1}{24}\cdot(-\tfrac{1}{2}) + \tfrac{1}{6}\cdot\tfrac{1}{12} + \tfrac{1}{2}\cdot 0\Big) = -\tfrac{1}{720}.$$

Laurent series and the Bernoulli numbers (cont.)

Writing $a_n = B_n/n!$ defines the *Bernoulli numbers*

$$\{B_0, B_1, B_2, \dots\} = \{1, -\tfrac{1}{2}, \tfrac{1}{6}, 0, -\tfrac{1}{30}, 0, \tfrac{1}{42}, \dots\},$$

and dividing g^{-1} by z,

$$\frac{1}{e^z - 1} = z^{-1} - \tfrac{1}{2} + \tfrac{1}{12} z - \tfrac{1}{720} z^3 + \cdots = z^{-1} \sum_{n=0}^{\infty} \frac{B_n}{n!} z^n. \tag{2.5.4}$$

The recurrence for the Bernoulli numbers (from Faulhaber's sum-of-powers formula) is

$$B_p = -\frac{1}{p+1} \sum_{j=0}^{p-1} B_j \binom{p+1}{j}. \tag{2.5.5}$$

Detour: Lovelace, Bernoulli numbers, and computation

Ada Lovelace's 1843 notes on Babbage's Analytical Engine include an algorithmic table for Bernoulli numbers. The computational point is exactly the recurrence just derived:

$$B_p = -\frac{1}{p+1} \sum_{j=0}^{p-1} \binom{p+1}{j} B_j.$$

Starting from $B_0 = 1$, every new value B_p depends only on earlier values $B_0, \dots, B_{p-1}$. Thus the recurrence is not merely a formal identity; it is an effective program for producing the coefficients in

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} z^n.$$

This is the historical bridge in the book: special-function constants, power-series coefficients, and algorithms enter the story together.

Detour: Lovelace, Bernoulli numbers, and computation (cont.)

The same numbers appear in Faulhaber's formula. With the convention $B_1 = -1/2$,

$$\sum_{k=1}^{n} k^p = \frac{1}{p+1} \sum_{j=0}^{p} (-1)^j \binom{p+1}{j} B_j \, n^{p+1-j}. \tag{2.5.6}$$

For example, $p = 1$ gives

$$\sum_{k=1}^{n} k = \frac{1}{2}\left(n^2 + n\right).$$

Computing the needed binomial coefficients is itself recursive:

$$\binom{p+1}{j} = \binom{p}{j-1} + \binom{p}{j},$$

the Pascal-triangle rule used by Lovelace's tabular algorithm.

The Laurent series of $\cot z$

From $\cos z = \frac{1}{2}(e^{iz} + e^{-iz})$ and $\sin z = \frac{1}{2i}(e^{iz} - e^{-iz})$,

$$\cot z = \frac{\cos z}{\sin z} = i\,\frac{e^{iz} + e^{-iz}}{e^{iz} - e^{-iz}}. \tag{2.5.7}$$

Splitting the fraction (note both terms keep the $+$ sign),

$$\cot z = i\,\frac{e^{iz}}{e^{iz} - e^{-iz}} + i\,\frac{e^{-iz}}{e^{iz} - e^{-iz}} = i\Big(\frac{1}{e^{2iz} - 1} + \frac{1}{1 - e^{-2iz}}\Big).$$

Using (2.5.4) for $1/(e^{z} - 1)$ with $z \mapsto \pm 2iz$ and combining,

$$\cot z = i\sum_{p=0}^{\infty} \frac{B_p}{p!}\Big[(2iz)^{p-1} - (-2iz)^{p-1}\Big].$$

The bracket vanishes when p is odd, because then $p-1$ is even; only the even Bernoulli indices $p = 2n$ remain. This gives

$$\cot z = z^{-1}\sum_{n=0}^{\infty} \frac{(-1)^n B_{2n}\, 2^{2n}}{(2n)!}\, z^{2n} = \frac{1}{z} - \frac{z}{3} - \frac{z^3}{45} - \cdots . \tag{2.5.8}$$

(Check: $n = 0$ gives $1/z$; $n = 1$ gives $-B_2\, 4/2! \cdot z = -z/3$.)

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Classifying singularities

A *singularity* is a point where f is not defined. The Laurent series tells which kind:

- **Removable**: $a_n = 0$ for all $n < 0$. Example: $\operatorname{sinc} z$, with $\operatorname{sinc} 0 := 1$.
- **Pole of order** p: $a_{-p} \neq 0$ and $a_n = 0$ for $n < -p$. A simple pole has $p = 1$ (e.g. $e^z/(1-z)$ at $z = 1$); a double pole has $p = 2$ (e.g. $e^z/(1-z)^2$).
- **Essential**: infinitely many negative-power terms. Example:

$$\sin\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{-(2n+1)},$$

  which is wild near the origin even though $\sin z$ is entire.

A function analytic except at isolated poles is called *meromorphic*; e.g. $1/\sin z$ is meromorphic with poles at $z = \pm n\pi$.

Classifying singularities (cont.)

Boundary language. A boundary point of a set S is a point that can be approached both from inside S and from outside S. An *open* set omits all its boundary points; a *closed* set contains them. This is why the ordinary interval $(0, 1)$ is open, $[0, 1]$ is closed, and the real line can be compactified by adding ideal endpoint(s) at infinity.

The Riemann sphere. Stereographic projection

$$z = \tan \tfrac{1}{2}\theta \, e^{i\phi}$$

maps the plane onto a sphere, adding a single point at infinity. This defines the *extended complex numbers* $\mathbb{C} \cup \{\infty\}$: the unit circle becomes the equator, the interior the southern hemisphere, and lines become circles passing through ∞. A neighborhood of ∞ on the sphere corresponds to the exterior $\{|z| > R\}$ of a large disk in the plane.

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The residue theorem

The coefficient a_{-1} in the Laurent series is the *residue* of f at z_0, written $\mathrm{Res}[f, z_0]$. Integrating the Laurent series term by term and using $\oint_\gamma (z - z_0)^n \, dz = 2\pi i \, \delta_{n,-1}$, only the $n = -1$ term survives:

$$\oint_\gamma f(z) \, dz = 2\pi i \, a_{-1} = 2\pi i \; \mathrm{Res}[f, z_0]. \tag{2.7.1}$$

For several poles, excluding each by a small loop (the function is analytic between them) gives the *residue theorem*:

$$\oint_\gamma f(z) \, dz = 2\pi i \sum_k \mathrm{Res}[f, z_k] \tag{2.7.2}$$

summed over all poles z_k inside γ.

Computing residues.

- Simple pole: $\operatorname{Res}[f, z_0] = \lim_{z\to z_0}(z-z_0)f(z)$.
- Double pole: $\operatorname{Res}[f, z_0] = \lim_{z\to z_0} \dfrac{d}{dz}\Big[(z-z_0)^2 f(z)\Big]$.
- Order-n pole: $\operatorname{Res}[f, z_0] = \lim_{z\to z_0} \dfrac{1}{(n-1)!}\dfrac{d^{n-1}}{dz^{n-1}}\Big[(z-z_0)^n f(z)\Big]$.
- Ratio with simple zero of g: $\operatorname{Res}\big[\frac{f}{g}, z_0\big] = \dfrac{f(z_0)}{g'(z_0)}$.
- At infinity: $\operatorname{Res}[f, \infty] = -\operatorname{Res}\big[\frac{1}{z^2}f(1/z), 0\big]$.

Examples.

$$\operatorname{Res}\Big[\tfrac{1}{z^2+1}, i\Big] = \frac{1}{2z}\Big|_{z=i} = \frac{1}{2i}, \qquad \operatorname{Res}\Big[\tfrac{1}{(z^2+1)^2}, i\Big] = \frac{-2}{(z+i)^3}\Big|_{z=i} = \frac{1}{4i}.$$

$$\operatorname{Res}\Big[\tfrac{\sin z}{z^2}, 0\Big] = \cos z\Big|_{z=0} = 1, \qquad \operatorname{Res}[z+1, \infty] = -\operatorname{Res}\Big[\tfrac{1}{z^3} + \tfrac{1}{z^2}, 0\Big] = 0.$$

The residue theorem (cont.)

Two quick contour integrals. On $|z| = 1$,

$$\oint \frac{\sin z}{z^2}\, dz = 2\pi i \ \mathrm{Res}\Big[\tfrac{\sin z}{z^2}, 0\Big].$$

Since $\sin z = z - \frac{z^3}{3!} + \cdots$, we have $\sin z/z^2 = z^{-1} - z/3! + \cdots$, so the residue is 1 and the integral is $2\pi i$. For the other example,

$$\frac{z^2}{\sin z} = \frac{z^2}{z - \frac{z^3}{3!} + \cdots} = \frac{z}{1 - \frac{z^2}{3!} + \cdots},$$

which is analytic at 0 after defining its value there to be 0. Hence its residue is 0, and $\displaystyle\oint_{|z|=1} \frac{z^2}{\sin z}\, dz = 0.$

Example ($z^2/(e^z - 1)$). The equation $e^z - 1 = 0$ gives $z = 2\pi i m$ for $m \in \mathbb{Z}$. At $m = 0$, however, the numerator cancels the zero:

$$\frac{z^2}{e^z - 1} = \frac{z^2}{z + \frac{1}{2}z^2 + \cdots} = \frac{z}{1 + \frac{1}{2}z + \cdots},$$

The residue theorem (cont.)

so $z = 0$ is removable and its residue is 0. For $m \neq 0$ the poles are simple, and using $g(z) = e^z - 1$, $g'(z) = e^z$, gives

$$\operatorname{Res}\left[\frac{z^2}{e^z - 1}, 2\pi im\right] = \frac{(2\pi im)^2}{e^{2\pi im}} = -4\pi^2 m^2.$$

Branch-point example and a $\sum 1/n^2$ via residues

Branch point. For $f(z) = \dfrac{z^{-k}}{z+1}$ with $0 < k < 1$, use the branch $\arg z \in (0, 2\pi)$, with the cut along the positive real axis. Then $z = -1$ has $\arg z = \pi$, so the simple pole at $z = -1$ has residue

$$\mathrm{Res}[f, -1] = z^{-k}\Big|_{z=-1} = (-1)^{-k} = (e^{i\pi})^{-k} = e^{-i\pi k}.$$

The point $z = 0$ is a branch point, not an isolated singularity, so it has no Laurent series there (the pole/essential classification does not apply). A small circle of radius ε around 0 contributes at most $2\pi\varepsilon \cdot \varepsilon^{-k}/(1-\varepsilon)$, which tends to 0 as $\varepsilon \to 0$; this is a vanishing small-loop contribution, not a residue at 0.

Basel sum $\sum_{n=1}^{\infty} \frac{1}{n^2}$**.** The function $f(z) = \frac{\pi \cot(\pi z)}{z^2}$ has poles at every integer. From the Laurent series of $\pi \cot \pi z$ about $z = n$,

$$\pi \cot \pi z = \frac{1}{z-n} - \frac{\pi^2 (z-n)}{3} - \frac{\pi^4 (z-n)^3}{45} - \cdots , \qquad (2.7.3)$$

one reads off $\operatorname{Res}[f, n] = 1/n^2$ for $n \neq 0$ and $\operatorname{Res}[f, 0] = -\pi^2/3$. On a square contour of side N (odd integer), the contour stays a distance at least $N/2$ from the origin and has length $4N$. Since $|\cot \pi z| < 2$ there, the ML-estimate gives

$$\left| \frac{1}{2\pi i} \oint f \, dz \right| \leq \frac{1}{2\pi} \cdot \frac{2\pi}{(N/2)^2} \cdot 4N = \frac{16}{N} \to 0.$$

Summing all residues to zero,

$$\left(\sum_{n=-\infty}^{-1} \tfrac{1}{n^2} \right) + \left(-\tfrac{\pi^2}{3} \right) + \left(\sum_{n=1}^{\infty} \tfrac{1}{n^2} \right) = 0 \;\Rightarrow\; \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

More generally $\sum_{n=1}^{\infty} \frac{1}{n^{2s}} = \frac{(2\pi)^{2s} |B_{2s}|}{2(2s)!}$, giving $\pi^4/90$ and $\pi^6/945$ for $s = 2, 3$.

Euler's Basel shortcut for the Basel problem

Euler's original route compares the zeros of $\sin \pi x$ with its Taylor coefficients. Using the infinite product

$$\sin \pi x = \pi x \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right),$$

the x^3 term comes from choosing one factor $-x^2/n^2$ and taking 1 from all the others:

$$\sin \pi x = \pi x \left(1 - x^2 \sum_{n=1}^{\infty} \frac{1}{n^2} + O(x^4)\right) = \pi x - \pi x^3 \sum_{n=1}^{\infty} \frac{1}{n^2} + O(x^5).$$

The Taylor series is

$$\sin \pi x = \pi x - \frac{\pi^3 x^3}{3!} + O(x^5).$$

Comparing the x^3 coefficients gives $-\pi \sum_{n=1}^{\infty} n^{-2} = -\pi^3/6$, hence

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

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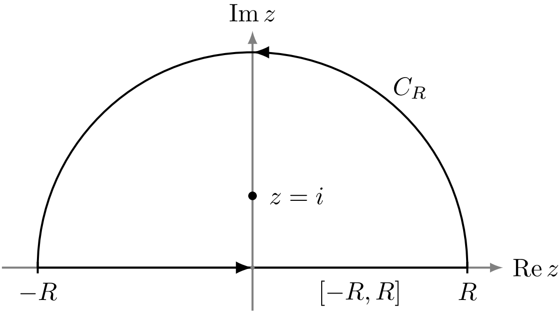
Real integrals by a semicircular contour

To evaluate $\int_{-\infty}^{\infty} f(x)\, dx$, close the contour with a large semicircle in the upper half-plane:



Real integrals by a semicircular contour (cont.)

$$\underbrace{\oint_\gamma f\,dz}_{①} = \underbrace{\int_{-R}^{R} f(x)\,dx}_{②} + \underbrace{\int_0^\pi f(Re^{i\theta})Re^{i\theta}\,i\,d\theta}_{③}.$$

If the arc ③ vanishes as $R \to \infty$, then $\int_{-\infty}^{\infty} f =$ ① $= 2\pi i \sum$(residues in upper half-plane). The arc is controlled by the *ML-estimate*

$$\left|\int_\gamma f(z)\,dz\right| \le ML, \qquad M = \max_{z\in\gamma}|f(z)|, \quad L = \text{length of } \gamma,$$

which here gives $|\int_{C_R} f(z)\,dz| \le \pi R \max_{C_R}|f| \to 0$ provided f decays faster than $1/R$.

Real integrals by a semicircular contour (cont.)

Example. $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x}{1+x^2}\,dx = \operatorname{Re}\int_{-\infty}^{\infty} \frac{e^{iz}}{1+z^2}\,dz$. Since $|e^{iz}| \le 1$ in the upper half-plane, the arc obeys $\left|\int_{C_R}\right| \le \frac{1}{R^2-1}\pi R \to 0$. The pole at $z = i$ gives

$$\oint \frac{e^{iz}}{1+z^2}\,dz = 2\pi i\,\frac{e^{iz}}{2z}\bigg|_{z=i} = 2\pi i\,\frac{e^{-1}}{2i} = \pi e^{-1},$$

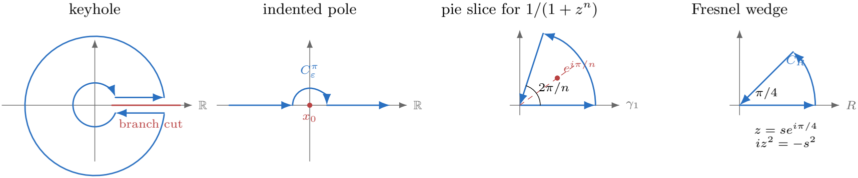
so $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x}{1+x^2}\,dx = \frac{\pi}{e}$.

Example. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{1+x^4}\,dx$. The roots of $1+z^4$ are $e^{i\pi/4}, e^{i3\pi/4}, \dots$; two lie in the upper half-plane. Summing residues $\dfrac{1}{4z^3}$ there,

$$\oint \frac{dz}{1+z^4} = 2\pi i\left(\tfrac{1}{4}e^{-i3\pi/4} + \tfrac{1}{4}e^{-i9\pi/4}\right) = \frac{\pi}{\sqrt{2}},$$

hence $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{1+x^4} = \frac{\pi}{\sqrt{2}}$.

Contour atlas used below



The keyhole contour compares the two sides of a branch cut; the indented contour isolates a real-axis pole; the pie-slice contour matches a rotated ray to the real-axis integral; the Fresnel wedge rotates e^{iz^2} into a Gaussian.

Keyhole contour: $\int_0^\infty x^a/(1+x)^2\,dx$

Evaluate $\displaystyle\int_0^\infty \frac{x^a}{(1+x)^2}\,dx$ for $-1<a<1$. First assume $a\neq 0$; the value at $a=0$ follows by continuity. The factor $z^a=|z|^a e^{ia\arg z}$ is multivalued, so use the branch $0<\arg z<2\pi$, cut along the positive real axis and use a *keyhole* contour. The integrand has a double pole at $z=-1$:

$$\oint_C \frac{z^a}{(1+z)^2}\,dz = 2\pi i\,\frac{d}{dz}(z^a)\Big|_{z=-1} = 2\pi i\,a\,e^{i\pi(a-1)} = -2\pi i\,a\,e^{i\pi a}.$$

The large circle has length $2\pi R$ and integrand size $O(R^a/R^2)=O(R^{a-2})$, so its contribution is $O(R^{a-1})\to 0$ because $a<1$. The small circle has length $2\pi\varepsilon$ and integrand size $O(\varepsilon^a)$, so its contribution is $O(\varepsilon^{a+1})\to 0$ because $a>-1$. Above the cut $z^a=|z|^a$; below it $z^a=|z|^a e^{i2\pi a}$, so the two straight pieces combine to $\left(1-e^{i2\pi a}\right)\int_0^\infty \frac{x^a}{(1+x)^2}\,dx$.

Keyhole contour: $\int_0^\infty x^a/(1+x)^2\,dx$ (cont.)

Therefore

$$\left(1 - e^{i2\pi a}\right) \int_0^\infty \frac{x^a}{(1+x)^2}\,dx = -2\pi i\, a\, e^{i\pi a}.$$

Dividing and writing $1 - e^{i2\pi a} = e^{i\pi a}(e^{-i\pi a} - e^{i\pi a}) = e^{i\pi a}(-2i \sin \pi a)$,

$$\int_0^\infty \frac{x^a}{(1+x)^2}\,dx = \frac{-2\pi i\, a\, e^{i\pi a}}{e^{i\pi a}(-2i \sin \pi a)} = \frac{\pi a}{\sin \pi a}. \tag{2.8.1}$$

The companion integral uses the same branch cut, but it is short enough to compute here. For $f(z) = z^{-a}/(1+z)$ with $0 < a < 1$, the pole at $z = -1$ has residue $(-1)^{-a} = e^{-i\pi a}$ on the chosen branch. The upper and lower straight pieces give $\left(1 - e^{-i2\pi a}\right) \int_0^\infty x^{-a}/(1+x)\,dx$. Hence

$$\left(1 - e^{-i2\pi a}\right) \int_0^\infty \frac{x^{-a}}{1+x}\,dx = 2\pi i\, e^{-i\pi a}.$$

Since $1 - e^{-i2\pi a} = e^{-i\pi a}(e^{i\pi a} - e^{-i\pi a}) = 2ie^{-i\pi a}\sin(\pi a)$,

$$\int_0^\infty \frac{x^{-a}}{1+x}\,dx = \frac{\pi}{\sin \pi a}, \qquad 0 < a < 1. \tag{2.8.2}$$

Cauchy principal value and a log integral

When a pole lies on the path of integration, the ordinary integral diverges. The *Cauchy principal value* lets both sides approach the pole at the same rate. For example, the two one-sided integrals of $1/x$ on $[-1,1]$ diverge, but

$$\mathrm{PV}\int_{-1}^{1}\frac{dx}{x}=\lim_{\varepsilon\to 0^+}\left(\int_{-1}^{-\varepsilon}\frac{dx}{x}+\int_{\varepsilon}^{1}\frac{dx}{x}\right)=0.$$

If the two sides are not synchronized the cancellation can change, so this is a separate notion from ordinary improper convergence:

$$\mathrm{PV}\int_{a}^{b}f(x)\,dx=\lim_{\varepsilon\to 0^+}\left(\int_{a}^{x_0-\varepsilon}+\int_{x_0+\varepsilon}^{b}\right)f(x)\,dx.$$

Indenting the contour by a small semicircle C_ε^θ of angle θ around a simple pole contributes

$$\int_{C_\varepsilon^\theta}f(z)\,dz\to i\sigma\theta\,\mathrm{Res}[f,z_0], \tag{2.8.3}$$

a fraction $\theta/2\pi$ of a full residue. Here $\sigma=+1$ for counterclockwise indentation and $\sigma=-1$ for clockwise indentation. For a half-circle, $\theta=\pi$.

Cauchy principal value and a log integral (cont.)

Example. $\displaystyle\int_0^\infty \frac{\log x}{x^2-1}\,dx$. Use $f(z)=\dfrac{\operatorname{Log} z}{z^2-1}$ on an upper semicircle indented at $z=-1,0,1$, with the principal boundary value $\operatorname{Log}(-1+i0)=i\pi$. The big and small arcs vanish. At $z=-1$ (a half-circle clockwise, $\theta=\pi$),

$$\int_{C_{-1}} \frac{\log z}{z^2-1}\,dz = -\pi i\,\frac{\log z}{2z}\Big|_{z=-1} = -\pi i\,\frac{i\pi}{-2} = -\frac{\pi^2}{2},$$

while at $z=1$ the residue $\log 1=0$ gives no contribution. Collecting real parts (the $\int_0^\infty$ piece is half the symmetric integral) yields

$$0=\mathrm{PV}\int_{-\infty}^{\infty}\frac{\operatorname{Log}(x+i0)}{x^2-1}\,dx+\left(-\frac{\pi^2}{2}\right)+0,$$

so

$$\mathrm{PV}\int_{-\infty}^{\infty}\frac{\operatorname{Log}(x+i0)}{x^2-1}\,dx=\frac{\pi^2}{2},\qquad \int_0^\infty\frac{\log x}{x^2-1}\,dx=\frac{\pi^2}{4}.$$

Worked example: a squared-log integral

Evaluate

$$I = \int_0^\infty \frac{(\log x)^2}{1+x^2}\, dx.$$

Use the principal logarithm in the upper half-plane, so $\operatorname{Log} x = \log x$ for $x > 0$ and $\operatorname{Log}(-x + i0) = \log x + i\pi$ for $x > 0$. Consider

$$F(z) = \frac{\operatorname{Log}^2 z}{1+z^2}$$

on the upper semicircle. The large arc vanishes because $|\operatorname{Log}^2 z/(1+z^2)| = O((\log R)^2/R^2)$ while the arc length is πR; the small indentation around 0 vanishes because $\varepsilon(\log \varepsilon)^2 \to 0$.

Worked example: a squared-log integral (cont.)

Only the pole at $z = i$ lies inside the contour. Since $1 + z^2 = (z - i)(z + i)$,

$$\operatorname{Res}[F, i] = \frac{\operatorname{Log}^2 i}{2i} = \frac{(i\pi/2)^2}{2i} = \frac{-\pi^2/4}{2i} = \frac{i\pi^2}{8}.$$

Therefore

$$\mathrm{PV}\int_{-\infty}^{\infty} \frac{\operatorname{Log}^2(x + i0)}{1 + x^2}\, dx = 2\pi i\, \frac{i\pi^2}{8} = -\frac{\pi^3}{4}.$$

Worked example: a squared-log integral (cont.)

Split the real-axis integral into positive and negative halves. On $x > 0$ the integrand is $(\log x)^2/(1+x^2)$. On $x < 0$, write $x = -t$:

$$\mathrm{Log}(-t + i0) = \log t + i\pi,$$

so the negative half contributes

$$\int_0^\infty \frac{(\log t + i\pi)^2}{1+t^2}\,dt = I + 2i\pi \int_0^\infty \frac{\log t}{1+t^2}\,dt - \pi^2 \int_0^\infty \frac{dt}{1+t^2}.$$

The middle integral is zero by the substitution $t = 1/u$, and the last integral is $\pi/2$. Thus the full contour value is

$$I + \left(I - \frac{\pi^3}{2}\right) = 2I - \frac{\pi^3}{2}.$$

Equating with $-\pi^3/4$ gives

$$I = \int_0^\infty \frac{(\log x)^2}{1+x^2}\,dx = \frac{\pi^3}{8}. \tag{2.8.4}$$

Jordan's lemma and $\int \sin x/x \, dx$

Theorem (Jordan's Lemma)

If f is analytic in the upper half-plane except at finitely many singularities and $|f| \to 0$ as $|z| \to \infty$ there, then $\lim_{R\to\infty} \int_{C_R} f(z)\, e^{iaz}\, dz = 0$ for $a > 0$.

Proof sketch. With $z = Re^{i\theta}$ and $M_R = \max_{C_R} |f|$,

$$\left| \int_{C_R} f\, e^{iaz}\, dz \right| \le M_R R \int_0^{\pi} e^{-aR\sin\theta}\, d\theta = 2M_R R \int_0^{\pi/2} e^{-aR\sin\theta}\, d\theta.$$

Using $\sin\theta \ge 2\theta/\pi$ on $[0, \pi/2]$ and integrating the exponential,

$$\le 2M_R R \left(\frac{-\pi}{2aR} \right) e^{-2aR\theta/\pi} \Big|_0^{\pi/2} = M_R\, \frac{\pi}{a} \left(1 - e^{-aR}\right) \to 0,$$

since $M_R \to 0$. $\square$

Jordan's lemma and $\int \sin x/x\,dx$ (cont.)

Example. $\displaystyle\int_{-\infty}^{\infty} \frac{\sin x}{x}\,dx$. Take the imaginary part of $\int \frac{e^{iz}}{z}\,dz$ on a semicircle indented at $z = 0$. The big arc vanishes by Jordan's lemma; the small indentation about the pole contributes, in the limit as $\varepsilon \to 0$,

$$\lim_{\varepsilon\to 0}\int_{C_\varepsilon} \frac{e^{iz}}{z}\,dz = -\pi i\ \mathrm{Res}\Big[\tfrac{e^{iz}}{z}, 0\Big] = -\pi i.$$

With no poles enclosed, the full contour is zero, so $\mathrm{PV}\int_{-\infty}^{\infty} \frac{e^{ix}}{x}\,dx = \pi i$, and taking the imaginary part,

$$\int_{-\infty}^{\infty} \frac{\sin x}{x}\,dx = \pi.$$

Pie-slice contour: $\int_0^\infty 1/(1+x^n)\,dx$

For an integer $n > 1$, the zeros of $1 + z^n$ are $z = e^{i(2k+1)\pi/n}$; only $z = e^{i\pi/n}$ lies inside a wedge of angle $2\pi/n$. The contour is the boundary of that wedge: γ_1 out along the real axis, arc C_R, and γ_2 back at angle $2\pi/n$. At the simple pole the residue is $1/(nz^{n-1})$. Since $z_0^n = e^{i\pi} = -1$ there, $z_0^{n-1} = z_0^n/z_0 = -e^{-i\pi/n}$, so $1/(nz_0^{n-1}) = -e^{i\pi/n}/n$ and

$$\oint_\gamma \frac{dz}{1+z^n} = 2\pi i\,\frac{1}{nz^{n-1}}\bigg|_{z=e^{i\pi/n}} = -\frac{2\pi i}{n}\,e^{i\pi/n}.$$

Pie-slice contour: $\int_0^\infty 1/(1+x^n)\,dx$ (cont.)

On the return ray $z = re^{i2\pi/n}$ gives $z^n = r^n$ and $dz = e^{i2\pi/n}\,dr$, so γ_2 reproduces the real integral times $-e^{i2\pi/n}$. With the arc vanishing,

$$\left(1 - e^{i2\pi/n}\right) \int_0^\infty \frac{dr}{1+r^n} = -\frac{2\pi i}{n}\, e^{i\pi/n}.$$

Dividing and writing the denominator as $1 - e^{i2\pi/n} = e^{i\pi/n}\left(e^{-i\pi/n} - e^{i\pi/n}\right)$, the factor $e^{i\pi/n}$ cancels the one in the numerator; then $e^{-i\pi/n} - e^{i\pi/n} = -2i\sin(\pi/n)$ by (2.2.4), so

$$\int_0^\infty \frac{dx}{1+x^n} = -\frac{2\pi i}{n}\,\frac{e^{i\pi/n}}{e^{i\pi/n}\left(e^{-i\pi/n} - e^{i\pi/n}\right)} = -\frac{2\pi i}{n}\,\frac{1}{-2i\sin(\pi/n)} = \frac{\pi/n}{\sin(\pi/n)}. \tag{2.8.5}$$

sinc transform, Gaussian, and Fresnel integrals

Fourier transform of sinc. Writing $\frac{\sin t}{t} = \frac{e^{it}-e^{-it}}{2it}$ and combining with $e^{i\xi t}$ gives

$$\frac{\sin t}{t} e^{i\xi t} = \frac{e^{i(\xi+1)t} - e^{i(\xi-1)t}}{2it}.$$

Jordan's lemma gives the principal-value contour rule $\int_{-\infty}^{\infty} e^{iat} t^{-1}\, dt = i\pi \operatorname{sgn}(a)$, so the transform is $\frac{\pi}{2}\left[\operatorname{sgn}(\xi+1) - \operatorname{sgn}(\xi-1)\right]$, with half-height at the jump points. Thus

$$\int_{-\infty}^{\infty} \frac{\sin t}{t}\, e^{i\xi t}\, dt = \begin{cases} \pi, & -1 < \xi < 1, \\ \pi/2, & |\xi| = 1, \\ 0, & |\xi| > 1. \end{cases} \tag{2.8.6}$$

Gaussian integral. Squaring and going to polar coordinates,

$$\left(\int_{-\infty}^{\infty} e^{-x^2} dx\right)^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\, dr\, d\theta = \pi,$$

so $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$ and, scaling $x \mapsto x/\sqrt{m}$,

$$\int_{-\infty}^{\infty} e^{-mx^2}\, dx = \sqrt{\pi/m}, \qquad m > 0. \tag{2.8.7}$$

sinc transform, Gaussian, and Fresnel integrals (cont.)

Fresnel integrals. Take the real and imaginary parts of $\int_0^\infty e^{iz^2}\,dz$ on a pie slice of angle $\pi/4$. On the outer arc $z = Re^{i\theta}$, $0 \le \theta \le \pi/4$, substitute $u = z^2$. Its image is a quarter-circle in the upper half u-plane, with the branch $\sqrt{u} = z$ on that arc and $dz = du/(2\sqrt{u})$. Since $|\sqrt{u}| = R$ there, Jordan's lemma gives

$$\left|\int_{C_R} e^{iz^2}\,dz\right| = \left|\int_{\Gamma_{R^2}} \frac{e^{iu}}{2\sqrt{u}}\,du\right| \le \frac{C}{R} \to 0,$$

where Γ_{R^2} is the image quarter-circle and the last estimate is Jordan's lemma on that arc, with C independent of R. Along the ray $z = e^{i\pi/4}s$ the integral reduces to a Gaussian:

$$\int_0^\infty e^{iz^2}\,dz = e^{i\pi/4}\int_0^\infty e^{-s^2}\,ds = e^{i\pi/4}\frac{\sqrt{\pi}}{2} = \frac{\sqrt{\pi}}{2}\left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right).$$

Hence

$$\int_0^\infty \cos(x^2)\,dx = \int_0^\infty \sin(x^2)\,dx = \frac{\sqrt{\pi}}{2\sqrt{2}}. \tag{2.8.8}$$

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The steepest-descent approximation

We estimate

$$I(s) = \int_a^b g(z)\, e^{sf(z)}\, dz, \qquad s \gg 1, \tag{2.9.1}$$

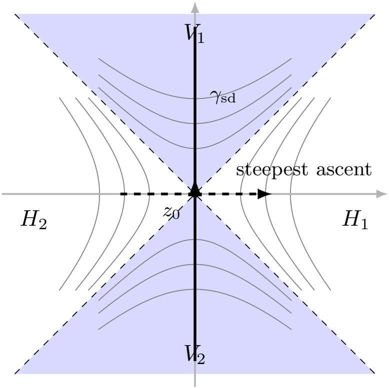
with f, g analytic. Write $f = u + iv$. At a saddle $f'(z_0) = 0$, so $u_x = v_x = 0$ at z_0; the Cauchy–Riemann equations then also give $u_y = -v_x = 0$ and $v_y = u_x = 0$. Thus both u and v have stationary first-order behavior there.

Differentiating the Cauchy–Riemann equations gives $u_{xx} + u_{yy} = 0$. The Hessian of $u = \operatorname{Re} f$ at z_0 has the form

$$D^2 u(z_0) = \begin{pmatrix} u_{xx} & u_{xy} \\ u_{xy} & -u_{xx} \end{pmatrix}, \qquad \det D^2 u(z_0) = -u_{xx}^2 - u_{xy}^2.$$

If this Hessian is nonzero, the determinant is negative, so u has a genuine saddle: increase in one direction and decrease in another.

The steepest-descent approximation (cont.)



The steepest-descent approximation (cont.)

Assume the contour can be deformed without crossing singularities or branch cuts to a nondegenerate saddle z_0 where $f'(z_0) = 0$ and $f''(z_0) \neq 0$, and that $\operatorname{Re} f$ decreases along the chosen path away from the saddle. Near z_0,

$$f(z) \approx f(z_0) + \tfrac{1}{2} f''(z_0)(z - z_0)^2.$$

Write $f''(z_0) = |f''(z_0)| e^{i\theta}$ with $\theta = \arg f''(z_0)$, and take the path $z = z_0 + re^{i\phi}$. Choosing the *steepest-descent* direction $\phi = \frac{1}{2}(\pi - \theta)$ turns the exponent into a real Gaussian:

$$e^{\frac{1}{2} s f''(z_0)(z - z_0)^2} = e^{\frac{1}{2} s |f''(z_0)| e^{i\theta} r^2 e^{i2\phi}} = e^{-\frac{1}{2} s |f''(z_0)| r^2}.$$

The steepest-descent approximation (cont.)

For $s \gg 1$ only the neighborhood of z_0 matters; set $g \approx g(z_0)$ and extend the Gaussian over all r. We parametrize the chosen descent direction by $z = z_0 + ie^{-i\theta/2}r$ with r increasing along the deformed contour, so $dz = ie^{-i\theta/2}dr$; reversing the contour would reverse this sign.

$$I(s) \approx g(z_0)e^{sf(z_0)}\, ie^{-i\theta/2} \int_{-\infty}^{\infty} e^{-\frac{1}{2}s|f''(z_0)|r^2}\, dr,$$

and using the Gaussian (2.8.7),

$$I(s) \approx g(z_0)\, e^{sf(z_0)}\, i\, e^{-i\theta/2}\left(\frac{2\pi}{s|f''(z_0)|}\right)^{1/2} = \frac{\sqrt{2\pi}\, g(z_0)\, e^{sf(z_0)}\, i\, e^{-i\theta/2}}{\sqrt{s|f''(z_0)|}}. \qquad (2.9.2)$$

Worked example: large binomial coefficient

Approximate $\dbinom{n}{m} = \dfrac{n!}{m!(n-m)!}$ for large n, with $0 < m < n$, $t = m/n \in (0,1)$, and both m and $n-m$ large. From the binomial expansion $(1+z)^n = \sum_k \binom{n}{k} z^k$, the coefficient is a contour integral:

$$\binom{n}{m} = \frac{1}{2\pi i} \oint \frac{(1+z)^n}{z^{m+1}}\, dz = \frac{1}{2\pi i} \oint \frac{1}{z}\, e^{n[\log(1+z) - t\log z]}\, dz, \qquad t = \frac{m}{n}.$$

So $f(z) = \log(1+z) - t\log z$. Its derivatives:

$$f'(z) = \frac{1}{1+z} - \frac{t}{z}, \qquad f''(z) = -\frac{1}{(1+z)^2} + \frac{t}{z^2}.$$

Setting $f'(z_0) = 0$ gives the saddle $z_0 = \dfrac{t}{1-t} = \dfrac{m}{n-m}$, real and positive, with $f''(z_0) = \dfrac{(1-t)^3}{t}$, so $\theta = \arg f''(z_0) = 0$.

Worked example: large binomial coefficient (cont.)

Here the path through the (real) saddle goes in the imaginary direction, $\phi = \frac{1}{2}\pi$, which contributes a phase $i = e^{i\pi/2}$. Applying (2.9.2) with $g(z) = 1/(2\pi i z)$,

$$\binom{n}{m} \approx \frac{1}{2\pi i}\frac{1}{z_0}\, e^{i\pi/2}\, e^{nf(z_0)}\left(\frac{2\pi}{nf''(z_0)}\right)^{1/2} = \frac{\exp[-n(t\log t + (1-t)\log(1-t))]}{\sqrt{2\pi n t(1-t)}}. \tag{2.9.3}$$

Equivalently $\binom{n}{m} \approx \frac{1}{\sqrt{2\pi}} \frac{n^{n+\frac{1}{2}}}{m^{m+\frac{1}{2}}(n-m)^{n-m+\frac{1}{2}}}$.

Numerical check. For $n = 88$, $m = 18$ the exact value is

$$\binom{88}{18} = 2418561960739869780 \approx 2.4186 \times 10^{18},$$

and (2.9.3) gives 2.4304×10^{18}, within half a percent.

Summary

- A function is **analytic** on an open neighborhood where its complex derivative exists. With continuous first partials on that neighborhood, the **Cauchy–Riemann conditions** are equivalent to complex differentiability; the resulting analytic function has a convergent **Taylor series** locally.
- Near a singularity the **Laurent series** applies; its a_{-1} coefficient is the **residue**.
- **Cauchy's theorem** ($\oint f = 0$ for analytic f) and the **residue theorem** ($\oint f = 2\pi i \sum \mathrm{Res}$) evaluate contour integrals by counting poles.
- Closing contours (semicircle, keyhole, pie-slice, indented) plus **Jordan's lemma** and the *ML*-**estimate** compute real integrals and infinite sums ($\sum 1/n^2 = \pi^2/6$, the Gaussian and Fresnel integrals, $\pi a/\sin \pi a$).
- The **method of steepest descent** gives large-parameter asymptotics, e.g. the binomial coefficient—the tool we carry into the study of the named special functions.